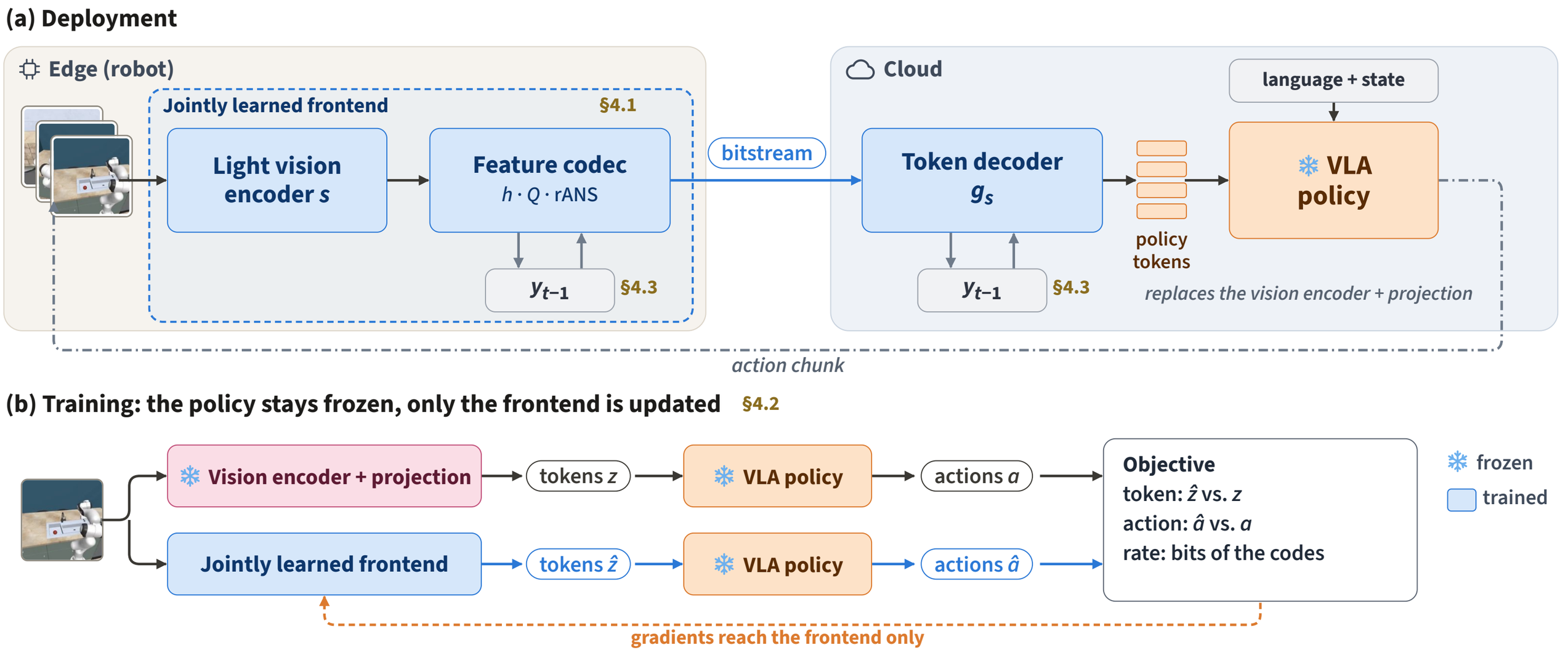

(a) Deployment
Edge (robot)
Jointly learned frontend
§4.1
Light vision encoder s
Feature codec
h · Q · rANS
y_{t−1}
§4.3
bitstream
Cloud
language + state
Token decoder g_s
y_{t−1}
§4.3
policy tokens
VLA policy
replaces the vision encoder + projection
action chunk
(b) Training: the policy stays frozen, only the frontend is updated
§4.2
Vision encoder + projection
tokens z
VLA policy
actions a
Jointly learned frontend
tokens ẑ
VLA policy
actions â
Objective
token: ẑ vs. z
action: â vs. a
rate: bits of the codes
frozen
trained
gradients reach the frontend only